

Processes

Definition

A **process** is a running program. Modern operating systems use processes to manage many activities happening apparently at the same time.

1. Why Processes Are Needed

Main Idea

Modern computers often do several things at once. To manage these activities properly, the operating system needs a way to model and control **concurrency**. Processes provide that model.

Concurrency

Concurrency means multiple activities are in progress during the same period of time, even if one CPU is actually executing only one instruction at any instant.

2. Example: Web Server

Web Server Example

A web server receives requests from many users asking for web pages.

What Happens

When a request arrives:

- the server checks whether the page is in cache,
- if it is, the page is sent back quickly,
- if it is not, a disk request is started.

Problem

Disk operations are very slow from the CPU's point of view. While the system is waiting for one disk request to finish, many more requests may arrive.

Need for Processes

So the server must handle multiple outstanding activities at the same time. Processes, and especially threads, help manage this concurrency.

3. Example: User PC

PC Example

A personal computer also performs many activities at once, often without the user noticing them.

Hidden Processes

After booting, the system may automatically start processes for:

- checking incoming email,
- updating antivirus definitions,
- printing files,
- backing up photos,
- running web browsing tasks.

Need for Management

All of these activities must be managed properly. Therefore, a **multiprogramming system** supporting multiple processes is very useful.

4. Multiprogramming

Definition

Multiprogramming is a system in which multiple processes are kept in memory and the CPU switches among them rapidly.

CPU Switching

In a multiprogramming system:

- the CPU runs one process for a short time,
- then switches to another process,
- then switches again, and so on.

Time Slice

Each process may run for only tens or hundreds of milliseconds before the CPU moves to another process.

5. Illusion of Parallelism

Important Idea

Strictly speaking, on a single-CPU system only **one process** runs at any exact instant. However, because the CPU switches so fast, it appears that many processes are running at the same time.

Pseudoparallelism

This apparent simultaneous execution is called **pseudoparallelism**.

Meaning

Pseudoparallelism gives the illusion of parallel work even though only one process is actually using the CPU at one instant.

6. True Parallelism

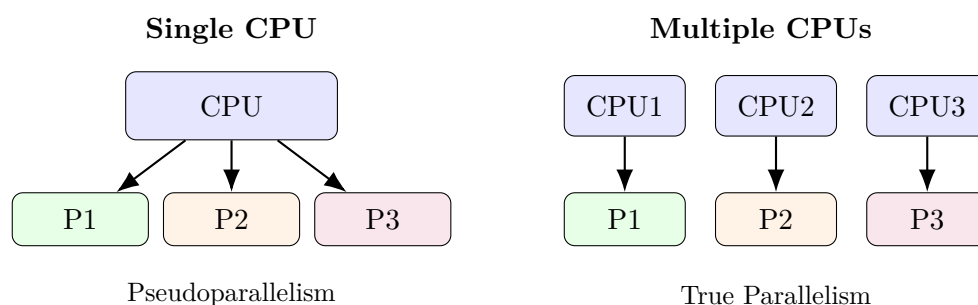
Real Hardware Parallelism

True parallelism occurs in multiprocessor systems, where two or more CPUs share the same physical memory and can actually run processes simultaneously.

Difference

- **Pseudoparallelism:** one CPU, rapid switching
- **True parallelism:** multiple CPUs, real simultaneous execution

6.1 Simple Comparison Diagram



7. Difficulty of Managing Parallel Activities

Human Difficulty

Keeping track of many parallel or seemingly parallel activities is difficult for people.

Why a Model is Needed

Because parallel activity is hard to understand directly, operating-system designers developed a conceptual model called **sequential processes**.

8. Sequential Process Model

Conceptual Model

A **sequential process** is a model in which each process is viewed as if it were executing its own sequence of instructions step by step.

Purpose

This model makes concurrency easier to understand, design, and manage.

9. Importance of Processes

Why Processes Matter

Processes are important because they help the operating system:

- organize multiple activities,
- share CPU time among tasks,
- handle concurrency,
- provide the illusion of several things happening together.

10. Summary Points

Quick Review

- A process is a running program.
- Modern computers handle many activities at once.
- Multiprogramming helps manage multiple processes.
- On one CPU, fast switching creates pseudoparallelism.
- On multiple CPUs, true parallelism is possible.
- The sequential-process model makes concurrency easier to handle.